Customer Churn Prediction

Using a Custom Shallow Neural Network

Shyam Vijay Jagani
(2611208)

Jasmi Alasapuri
(2571395)

Fatema Doctor
(2604383)

Parth Rathwa
(2509367)

Problem Definition & Dataset

Project Objective

The Business Mission

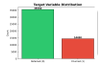
Develop an automated system to predict and prevent customer churn for an e-commerce subscription service.

Dataset Overview

- **Volume:** 50,000 customers
- **Features:** 25 Behavioral & Demographic
- **Churn Rate:** 28.9% (Imbalanced Class)

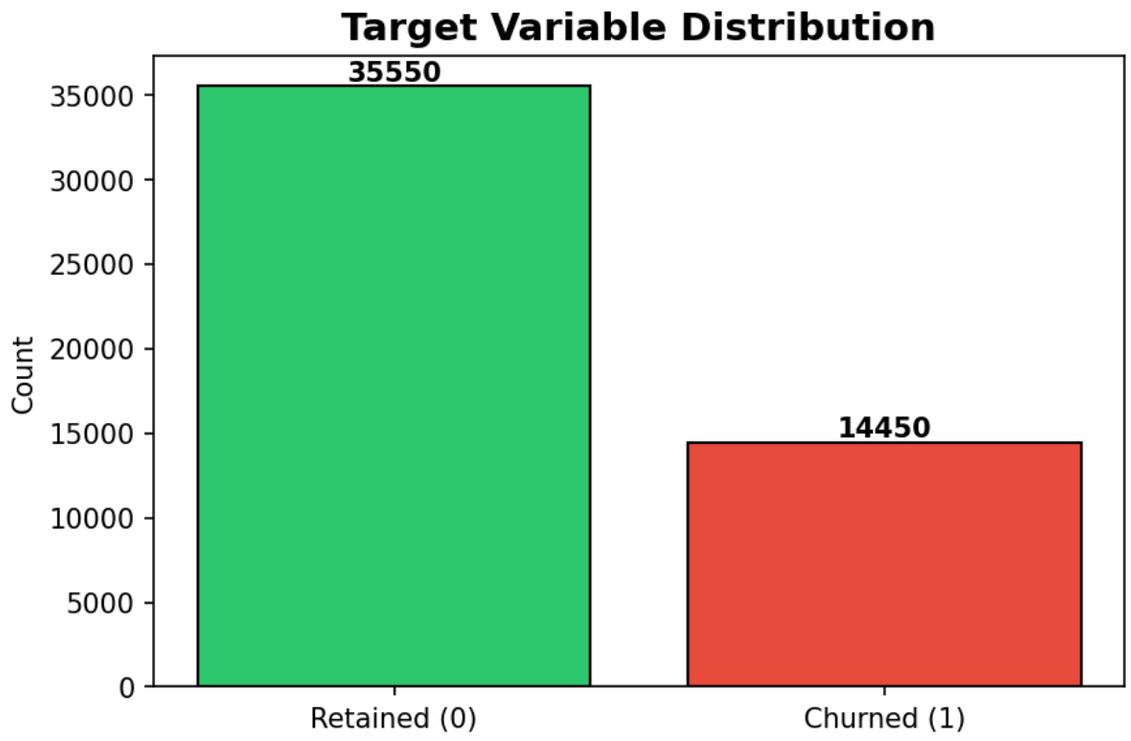
Core Challenges

1. **Complexity**
   - Non-linear relations
2. **Imbalance**
   - Favouring majority class
3. **Noisy Data**
   - High dimensionality



Exploratory Data Analysis

Target Distribution



Target Distribution

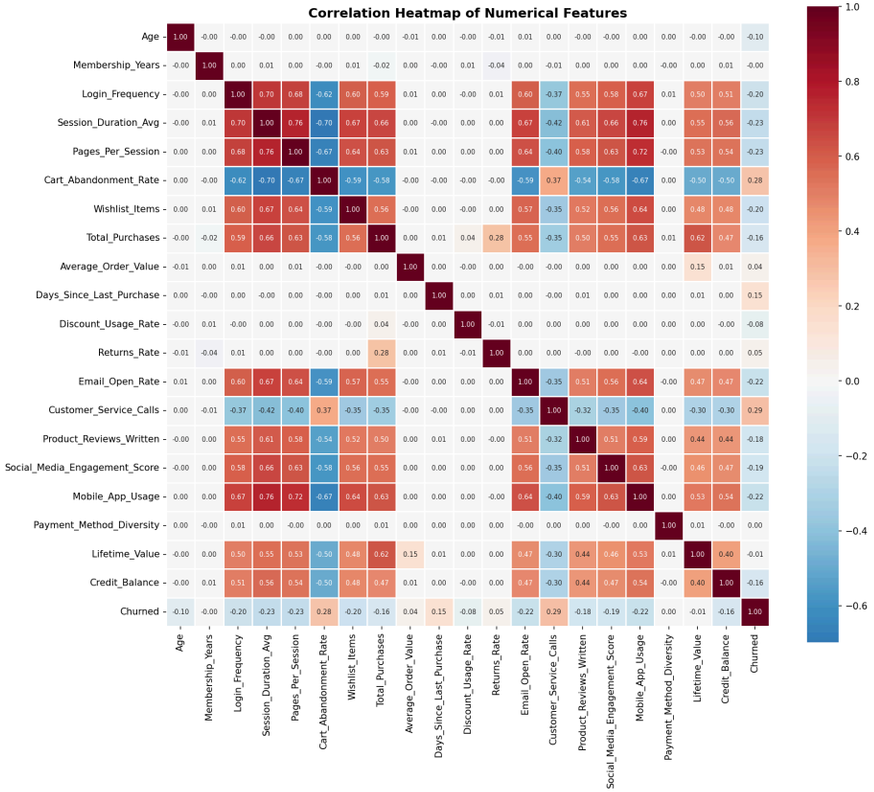
Observations

- **Churned:** 14,450
- **Retained:** 35,550

Implication

A standard accuracy metric would be misleading. We prioritize **AUC-ROC** and **Weighted Loss** to ensure we capture the minority churn class.

Correlation Insights



Correlation Heatmap

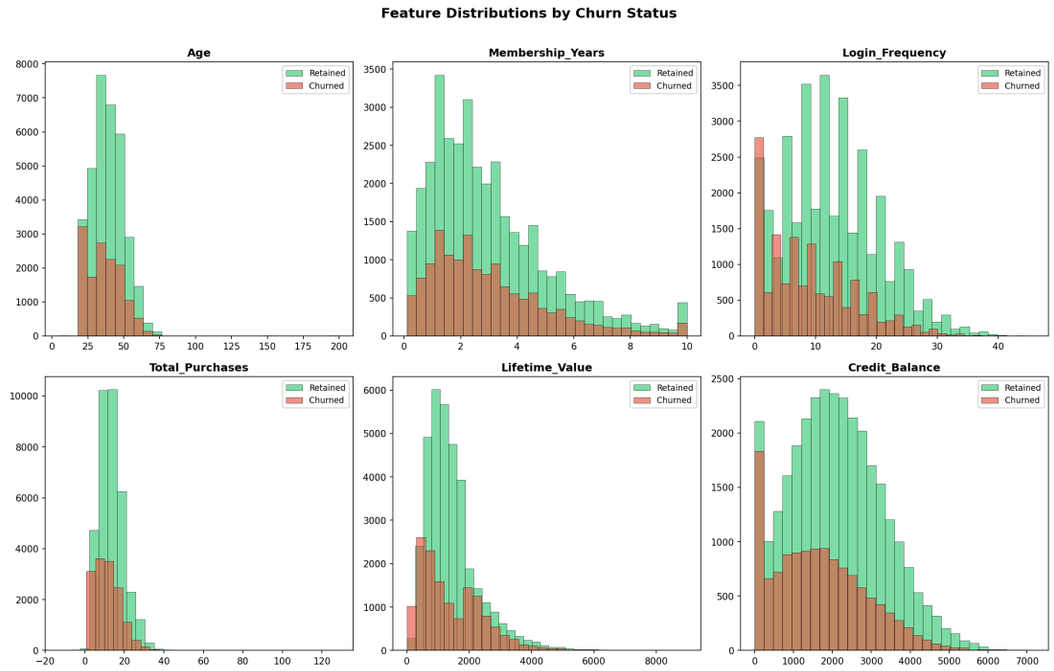
Primary Drivers

- `Days_Since_Last_Purchase` (+)
- `Cart_Abandonment_Rate` (+)

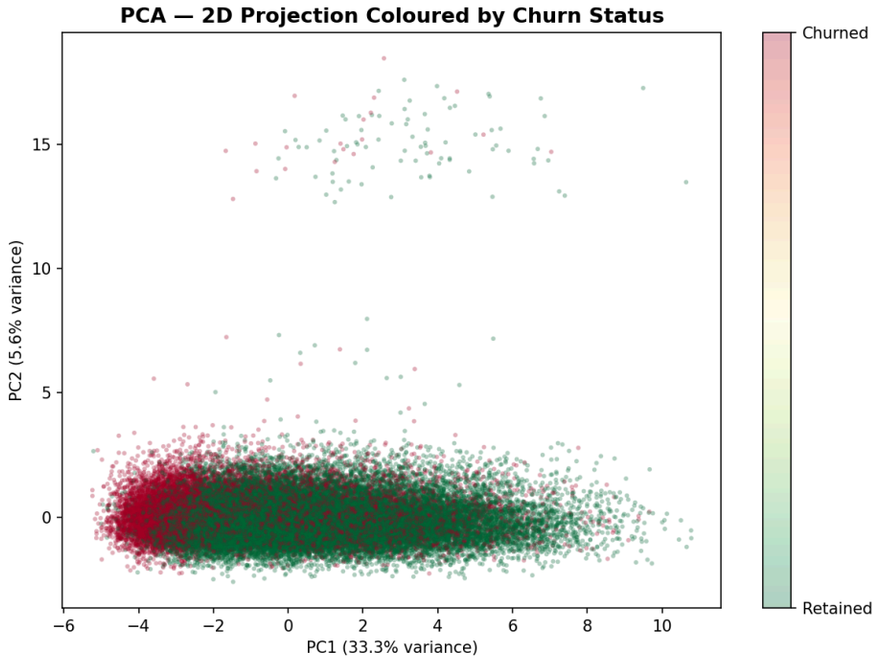
Protective Factors

- `Lifetime_Value` (-)
- `Membership_Years` (-)

Behavior & PCA



Distributions



PCA Scatter

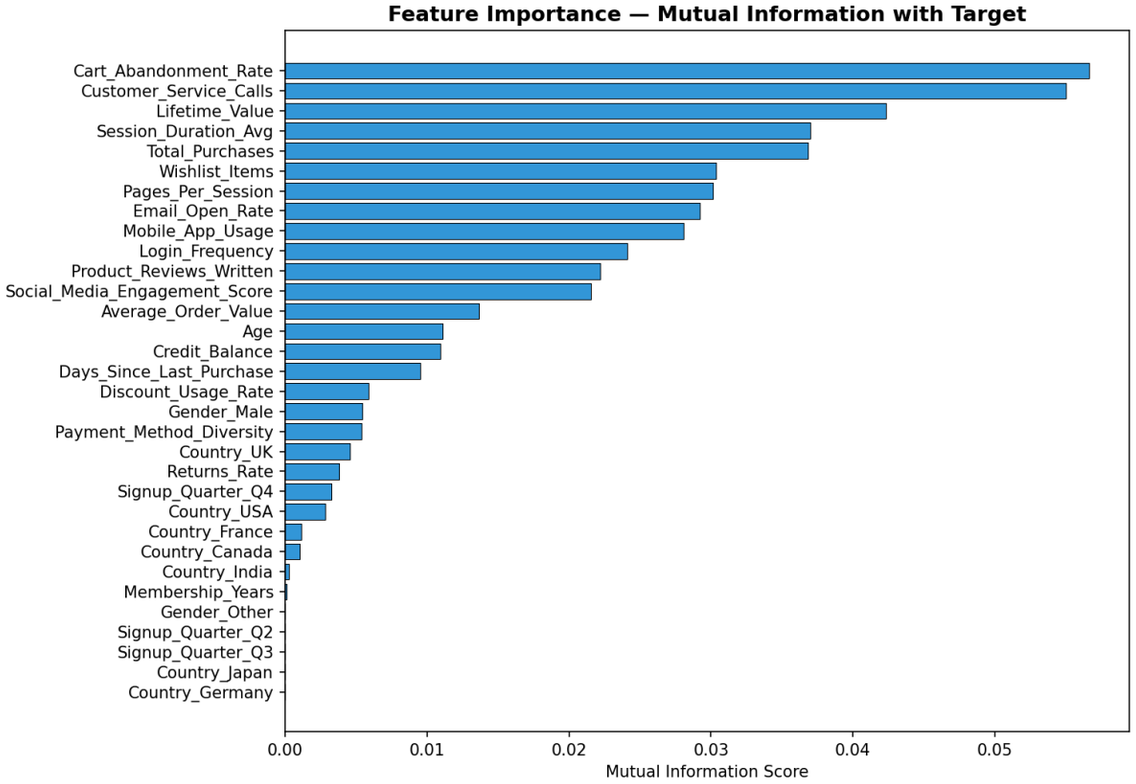
Insight: Overlapping distributions in PCA space justify the use of a **Neural Network** with non-linear activation to find complex boundaries.

Methodology & Engineering

Feature Selection Pipeline

Preprocessing

1. **Outlier Handling:** Winsorization
2. **Imputation:** Mode/Median
3. **Scaling:** StandardScaler



Mutual Information

MI Selection: Filtered 32 down to **25 high-influence features.**

Architecture: Custom NumPy SNN

Network Parameters

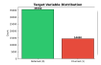
- **Input:** 25 Features
- **Hidden Layer:** 32 Units with **ReLU**
- **Output:** 1 Unit with **Sigmoid**

Implementation Detail

Built from scratch using **NumPy** for raw matrix control. Includes: - Manual Gradient Tracking via Chain Rule. - **Weighted BCE Loss** to handle class imbalance. - **He Initialization** for stable ReLU activation.

Architecture: Design Decisions

Component	Choice	Rationale
Hidden Units	32	Optimal capacity balance
Activation	ReLU	Prevents vanishing gradients
Loss	Weighted BCE	Handles 28% churn rate
Regularization	L2 (λ=0.001)	Penalizes complex weights
Early Stopping	Patience=30	Stops at epoch ~195



Training & Results

Optimization Strategy

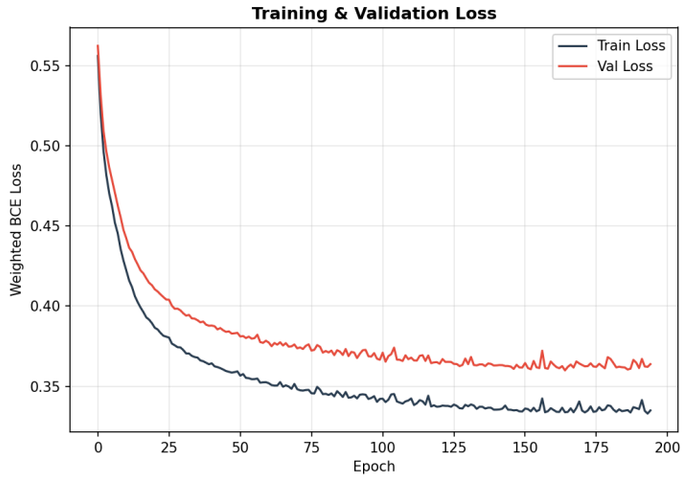
Hyperparameter Search

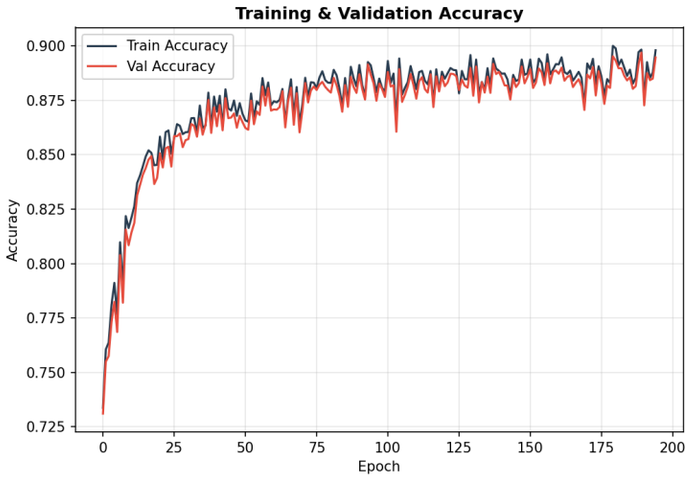
Tested LR (0.01-0.05) and L2 Lambda across 27 configurations.

Best Configuration

- **Learning Rate:** 0.05
- **L2 Lambda:** 0.001
- **Validation F1:** 0.81
- **Validation AUC:** 0.91

Training Stability

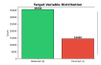




Training Curves

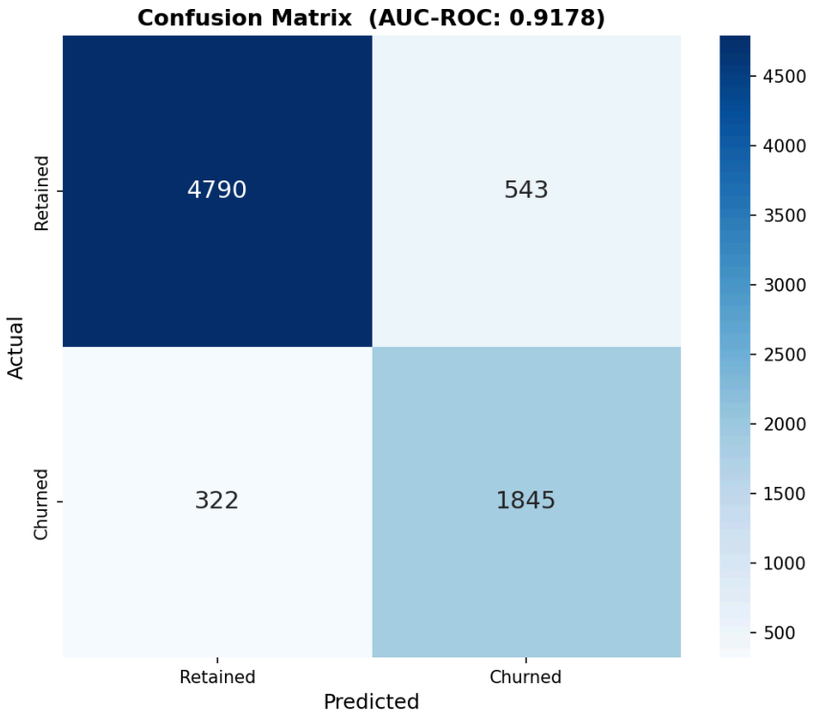
Convergence

- **Epochs:** 500 (Stopping at 195)
- **Validation Loss:** 0.3599
- Minimal gap between Train/Val suggests robust generalization without overfitting.

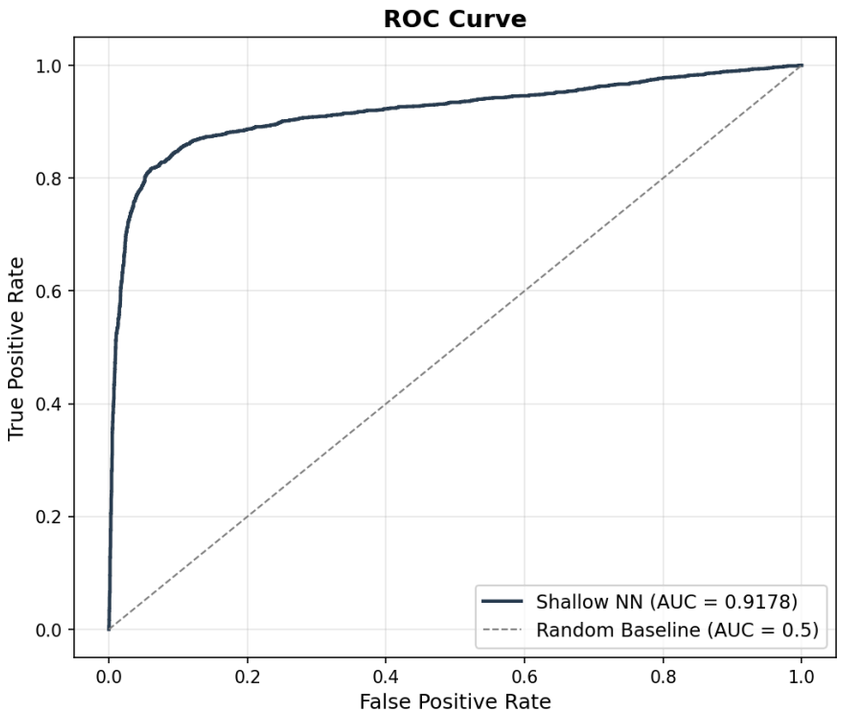


Final Performance

Evaluation Metrics



Confusion Matrix

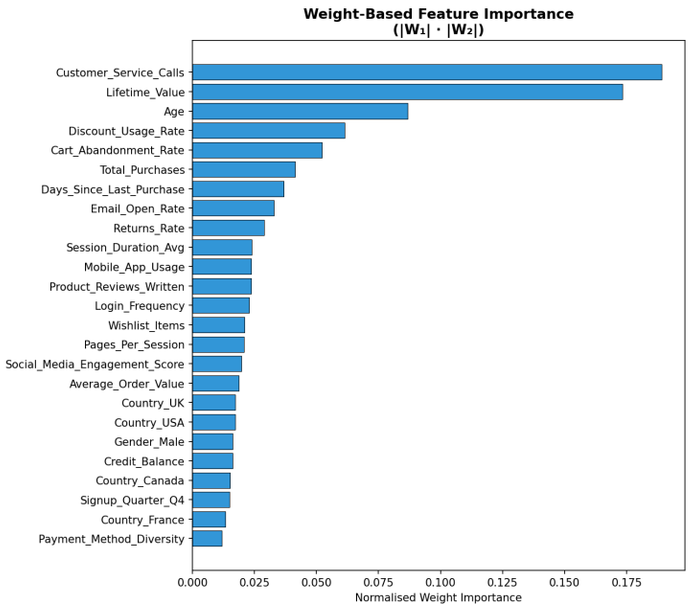


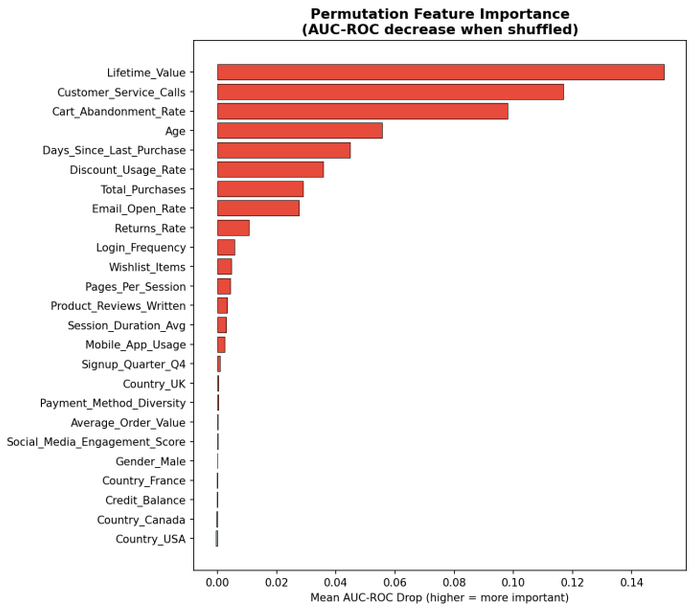
ROC Curve

Metric	Score
AUC-ROC	**0.91**
Precision	0.81
Recall	0.81

Business Impact

Feature Importance (The Why)

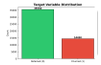




Critical Churn Indicators

1. **Service Friction:** High number of support calls.
2. **Engagement Drop:** High Cart Abandonment.
3. **Inactivity:** Days since last purchase.

Feature Importance



Retention Action Plan

Retention Strategy

We recommend targeted intervention for customers with >3 service calls.

- **Priority Outreach:** Flag high-friction service callers.
- **Win-Back Flows:** Auto-trigger for customers past their purchase window.
- **Loyalty Incentives:** Focus on customers in their first 2 years of membership.

Conclusion

Summary

- Custom NumPy Neural Network
- High AUC-ROC (0.91)
- Balanced Precision/Recall
- Explainable insights for Business Strategy

Q&A

Thank You!